

Puluputturi Nithin Reddy

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Professional Summary

Software Systems Engineer and Published AI Researcher with expertise in **distributed consensus (Raft)**, **low-level storage engines (C++ LSM-Trees)**, and **edge computer vision / test-time adaptation**. Published author in **IEEE and Springer Nature**, bridging systems engineering with production full-stack services (React, Spring Boot, .NET Core).

Education

Amrita Vishwa Vidyapeetham <i>Bachelor of Technology in Computer Science and Engineering</i>	Oct 2022 – June 2026 GPA: 7.77 / 10.0
Narayana Junior College <i>Higher Secondary Certificate (Class XII) – MPC Stream</i>	June 2020 – June 2022 Percentage: 94.4%

Professional Experience

Psiog Digital <i>Software Engineer – Custom Applications Domain</i>	Chennai, India Aug 2026 – Present
- Architect and deploy full-stack custom enterprise modules using React 18 on the frontend and Spring Boot microservices on the backend. - Implement resilient RESTful API contracts and relational database persistence workflows handling high-concurrency client requests. - Author automated unit and integration tests (Jest, JUnit) ensuring continuous test coverage and software reliability across production releases.	
<i>Graduate Engineer Trainee (Intern)</i>	Feb 2026 – July 2026
- Built modular backend services in C# and .NET Core utilizing Object-Oriented Design Principles (OOP) and Microsoft SQL Server. - Formulated optimized database queries, stored procedures, and schema migrations for enterprise customer systems.	

Research & Publications

- “YOLOv8-Driven Spatio-Temporal Framework for Real-Time Detection of Risk-Prone Patient Movements in Clinical Environments,” **IEEE SCEECS 2026**. [\[IEEE Xplore\]](#)
- “YOLOv8-Based Real-Time Patient Safety Monitoring with Temporal Analysis and Automated Alerts,” **IEEE INSTCON 2026**. [\[IEEE Xplore\]](#)
- “Advancing Sustainable Seaweed Production: Linking Technology, Policy, and Ecology for a Resilient Blue Economy,” **Springer Nature – Environmental Science and Pollution Research (ESPR)**, 2026. [\[SpringerLink\]](#)

Featured Projects

LSM-Tree Embedded Storage Engine (Modern C++20)	[GitHub]
- Engineered an embedded persistent key-value storage engine in modern C++20 based on Log-Structured Merge-trees (LSM). - Implemented append-only Write-Ahead Logging (WAL) for crash recovery, in-memory SkipList MemTable, and binary-indexed SSTables. - Integrated in-memory Bloom filters to eliminate redundant disk seeks and a tiered background Compaction engine for tombstone reclamation.	
Raft Distributed Fault-Tolerant Key-Value Store	[GitHub]
- Implemented the Raft consensus algorithm from scratch covering randomized leader election, heartbeat RPCs, and log replication. - Engineered a chaos test harness simulating network partitions and packet drops, verifying split-brain avoidance and linearizable consistency. - Built an atomic state machine supporting durable Write-Ahead Logging (WAL) and Compare-And-Swap (CAS) transactions.	
PatientMonitoringYOLO: Edge Clinical Movement & Fall Detection	[GitHub]
- Built an edge CV framework coupling YOLOv8 with rolling centroid Euclidean velocity achieving an F1-score of 0.947. - Engineered room context heuristics autonomously suppressing alarms when healthcare workers are present ($N \geq 2$), backed by 2 IEEE papers. - Built automated CI/CD validation pipelines with 100% test coverage across tracking heuristics and state transitions.	

Clinical-TTA-Edge: Test-Time Adaptation for Edge Vision Models

[\[GitHub\]](#)

- Architected an unsupervised Test-Time Adaptation (TTA) engine for edge YOLOv8 detectors to resolve clinical domain shift (night illumination, blur, low contrast) without target-domain labels or cloud retraining.
- Formulated spatially-structured detection entropy (TENT) and SHOT information maximization adapting only **10,592 normalization affine parameters (99.66% model frozen)**, preventing representation collapse.
- Engineered an entropy-gated streaming pipeline ($\tau = 0.38$) achieving **50%–80% compute savings** on in-distribution frames while adapting on shifted video in real-time, validated with automated test suites.

Technical Skills

Languages: C++20, Python, Java, C#, SQL, TypeScript, JavaScript

Systems & Distributed: Distributed Consensus (Raft), LSM-Tree, WAL, SSTables, Bloom Filters, Compaction, CMake

Backend & Frameworks: Spring Boot, .NET Core, ASP.NET Web API, Flask, Entity Framework Core, Raw ADO.NET

Frontend: React 18, Vite, Material-UI (MUI), HTML5, CSS3

AI/ML & Computer Vision: Test-Time Adaptation (TTA), TENT, SHOT, YOLOv8, PyTorch, OpenCV, Scikit-Learn, Temporal Dynamics

Databases & DevOps: Microsoft SQL Server, MySQL, Cloud Firestore, Firebase Auth, Git, Docker, CI/CD Actions, Pytest

Honors & Co-Curricular

- **UNESCO Chair Certified:** Experiential Learning and Sustainable Innovation.
- **Live-in-Labs Research:** Collaborated on deployable rural technology solutions for community sustainability.
- **ASCI Technical Association:** Designer and contributor for the departmental technical newsletter.